

Lecture 02:

Differences between distributions

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Textbook:

Press, Teukolsky, Vetterling, Flannery (2002)

Numerical Recipes in C++, Cambridge University Press.

B Efron & T Hastie (2016) Computer Age Statistical Inference,

Cambridge University Press.

Abstract:

We introduce further basic ideas, such as standard deviation and **standard error**, and the possibility of obtaining more friendly distributions by **transforming samples**.

We take an excursion into a statistical utopia to better understand **statistical reasoning**.

Returning to the present, we introduce the basic template for **hypothesis testing** and try out various ways of **assessing statistical differences** between distributions.

Overview:

1. Bessel's correction and standard error
2. Useful transformations
3. Student's t-distribution
4. Hypothesis testing
5. Student's t-test for different mean
6. F-Test for different variance

7.1. Testing different variances

7. Chi-square test for different distributions

8. Kolmogorov-Smirnov (KS) test for different distributions

1. Bessel's correction and standard error

How we compute variance depends on whether we know the true population mean, in which case we can compute **true variance**, or whether we have only a sample of data points, so that must be content to **estimate variance**.

Given n observations x_i and the **true population mean** μ , the **true variance** is

$$\text{Var}(x) = \frac{1}{n} \sum_i (x_i - \mu)^2$$

Given n observations x_i , but no further information about the population, the best estimate of variance is the **corrected sample variance**

$$m = \frac{1}{n} \sum_i x_i, \quad s^2 = \frac{1}{n-1} \sum_i (x_i - m)^2$$

This is easy to prove when our samples come from a normal distribution

$$x_i \sim \mathcal{N}(\mu, \sigma)$$

In this case, the expected variance from n samples evaluates as follows

$$\begin{aligned} \left\langle \frac{1}{n} \sum_i (x_i - m)^2 \right\rangle &= \frac{1}{n} \sum_i \langle (x_i - m)^2 \rangle = \langle x_i^2 - 2m x_i + m^2 \rangle = \\ &= \left\langle x_i^2 - \frac{2}{n} \sum_j x_i x_j + \frac{1}{n^2} \left(\sum_j x_j \right)^2 \right\rangle = \\ &= \frac{n-2}{n} \langle x_i^2 \rangle - \frac{2}{n} \sum_{j \neq i} \langle x_i \rangle \langle x_j \rangle + \frac{1}{n^2} \sum_j \left(\langle x_j^2 \rangle + \sum_{k \neq j} \langle x_k \rangle \langle x_j \rangle \right) \\ &= \frac{n-1}{n} (\sigma^2 + \mu^2) - \frac{n-1}{n} \mu^2 = \\ &= \frac{n-1}{n} \sigma^2 \end{aligned}$$

In other words, the best estimate for true variance is

$$\sigma^2 = \frac{1}{n-1} \sum_i (x_i - m)^2$$

as (for normally distributed samples) the expected value of this expression **equals** the true variance.

This insight is known as "Bessel's correction".

Standard error

Suppose we take n samples from a normal distribution

$$x_i \sim \mathcal{N}(\mu, \sigma), \quad i \in \{1, \dots, n\}$$

and imagine that we repeat this many times. Imagining further that we are careful to take new (hypothetical) samples every time, so that all are **independent**.

How will the **sample average**, m

$$m = \frac{1}{n} \sum_i x_i$$

be distributed over repetitions? As we have seen in the last lecture, m will also be distributed normally, with the same mean as the individual samples, but with a smaller variance:

$$m \sim \mathcal{N}(\mu_{ave}, \sigma_{ave}^2), \quad \mu_{ave} = \mu, \quad \sigma_{ave}^2 = \frac{\sigma^2}{n}$$

The standard deviation of individual samples and (hypothetical) sample means are related as

$$\text{Std}(x) = \sigma, \quad \text{Std}(m) = \frac{\sigma}{\sqrt{n}}$$

Perhaps confusingly, the latter is often called **standard error**:

$$\text{SE}(x) = \frac{\text{Std}(x)}{\sqrt{n}}$$

The standard error estimates the standard deviation of the **sample mean**. Its value is **inferred, not observed**, because it extrapolates to a hypothetical scenario in which we can repeat our experiment and collect new samples many times.

When we estimate the variance from the sample (using Bessel's correction), the standard error becomes

$$\text{SE}(x) = \frac{\text{Std}(x)}{\sqrt{n}} = \frac{1}{\sqrt{n}} \sqrt{\frac{1}{n-1} \sum_i (x_i - m)^2} = \sqrt{\frac{1}{n(n-1)} \sum_i (x_i - m)^2}$$

where m is the observed sample mean.

Summary

- The true variance of a set of samples can be computed only when the true mean μ of the underlying distribution is known

$$\sigma^2 = \frac{1}{n} \sum_i (x_i - \mu)^2$$

- When both mean and variance are estimated from the same samples, the best estimate for variance is

$$\sigma^2 = \frac{1}{n-1} \sum_i (x_i - m)^2, \quad m = \frac{1}{n} \sum_i x_i$$

- If the underlying distribution is normal, the expected value of this estimate equals the true variance.
- The **standard error** is the expected standard deviation of the sample mean. It tells us how we would expect the sample mean to vary, if we could repeat the experiment.

$$\text{SE}(x) = \sqrt{\frac{1}{n(n-1)} \sum_i (x_i - m)^2}$$

- Standard errors are typically used for drawing '**error bars**'.

2. Useful transformations

Z-scoring

When we compare different types of measurements (ion concentrations, blood values, spike rates, BOLD signals, EEG scalp potentials, and so on), the numerical scales often differ greatly.

For purposes of statistical comparison, it is therefore helpful to normalize the data points by subtracting the mean μ and dividing by the standard deviation σ

$$z_i = \frac{x_i - \mu}{\sigma}$$

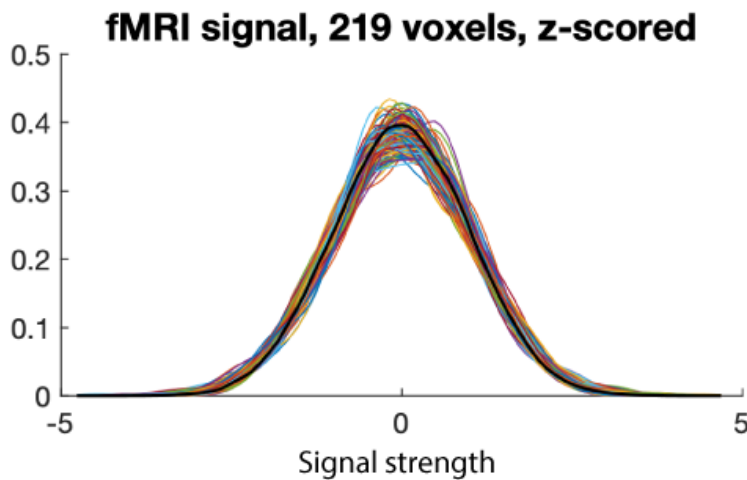
The transformed values are called '**z-scores**'. A side benefit of z-scoring is that it becomes more readily apparent if observations deviate from normality, are unstable over time, or exhibit other defects.

Obviously, when x is normally distributed, then z is standard normally distributed:

$$x \sim \mathcal{N}(\mu, \sigma) \quad \Rightarrow \quad z = \frac{x - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

σ

For example, z-scoring is applied routinely (after detrending and motion correction) to fMRI signals, which may not show the same standard deviation in all parts of the brain, because the field gradients do not image all parts equally well. Mean μ and standard deviation σ are obtained directly from all values recorded for one voxel.



Distribution of signal strength, after z-scoring, for 219 voxels in inferior temporal cortex.

Fisher's z-transform (optional)

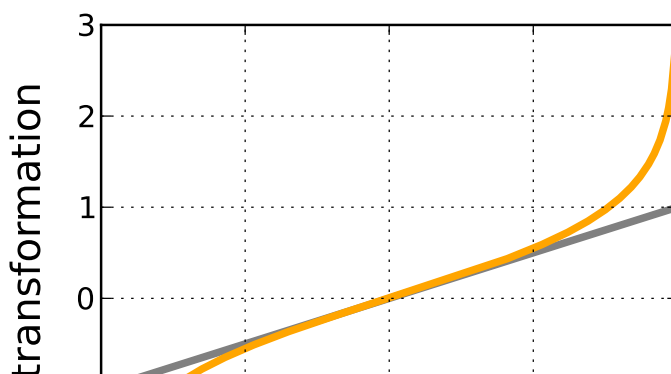
To avoid later confusion due to similar names, we mention also Fisher's z-transform, which serves a very different purpose.

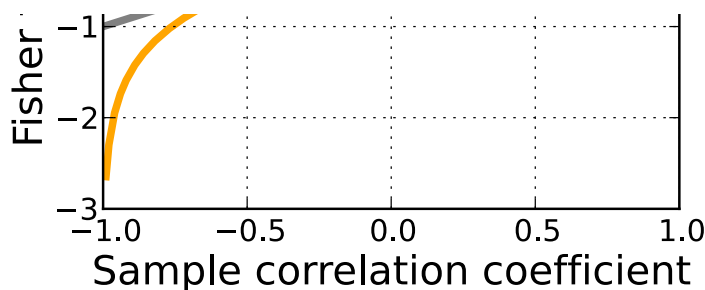
This maps values $r \in [-1, +1]$ ranging from $r = -1$ to $r = +1$ to values $z \in [-\infty, +\infty]$ ranging over the entire real axis.

Fisher's z-transform is defined as the inverse hyperbolic tangent

$$z = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right) = \tanh^{-1}(r)$$

The standard application is to transform correlation coefficients such as to facilitate statistical comparison. However, the z-transform can also be useful in other context.





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✓ Logarithmic transformation of positive data >

As mentioned, many biological observations fall naturally on a geometrical scale, with multiplicative rather than additive increments.

When such observations are transformed logarithmically, their distribution may become approximately normal, facilitating statistical comparison.

A logarithmic transformation requires that the data points are restricted to positive values, $x \in \mathbb{R}^+$.

Transformed values y are obtained from observed values x with

$$y = \frac{1}{\sigma} \ln\left(\frac{x}{\mu}\right)$$

Given the mean μ_x and variance σ_x^2 of our original observations x , we choose

$$\mu = \left(\frac{\mu_X^2}{\sqrt{\mu_X^2 + \sigma_X^2}} \right), \quad \sigma^2 = \ln\left(1 + \frac{\sigma_X^2}{\mu_X^2} \right)$$

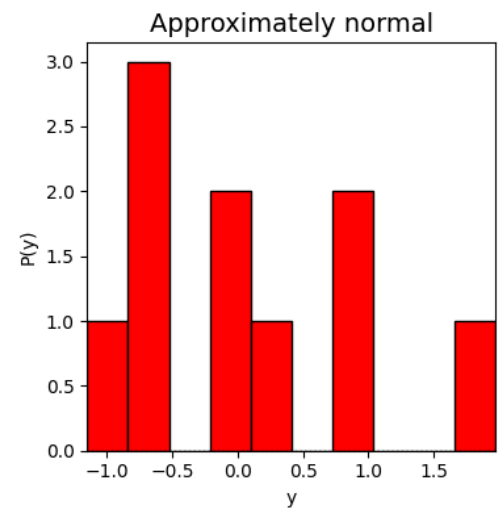
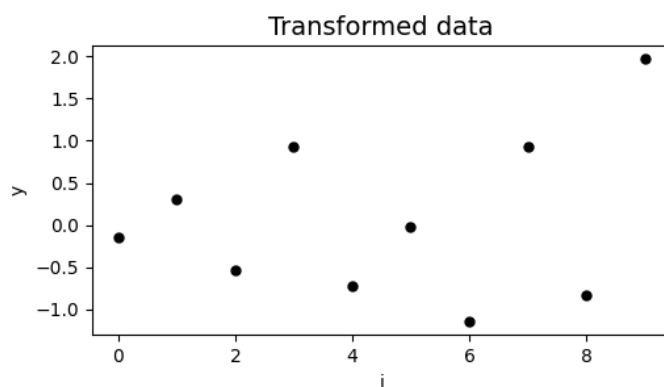
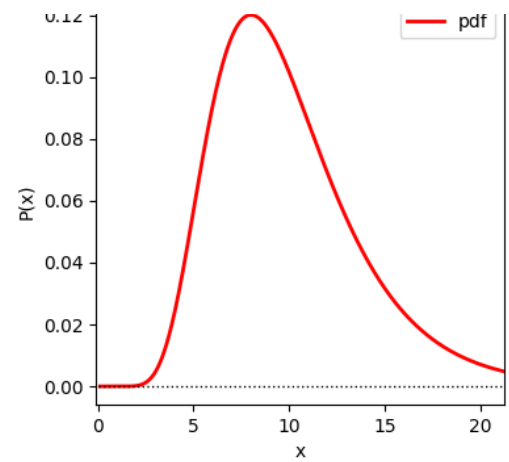
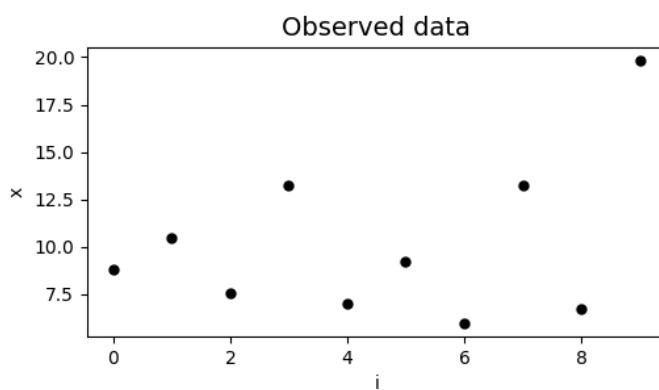
as parameters for the transformation.

Logarithmic transformation

Show code

muX 10
 sigmaX 4
 N 10

Lognormal distribution



```
plot_logarithmic_transformation
```

```
def plot_logarithmic_transformation(muX=10, sigmaX=4, N=10)
```

(Slide-25)

3. Student's t-distribution

We are trying to estimate the mean of n samples x_i , which we believe to stem from a normal distribution, but unfortunately the number n is rather small.

How confident can we be about our sample mean? How far from the unknown true mean can

How confident can we be about our sample mean: How far from the unknown true mean can we expect it to be?

$$m_{obs} = \frac{1}{n} \sum_i x_i$$

To address this question, let's enter into a hypothetical world where we know everything and can do anything we want. Surely you can see the attraction of such an escapist fantasy world!



William Gosset, who published his famous test under the pseudonym 'Student', was presumably a practical man, as he held the important position of "Chief Brewer" at Guinness Brewery in Dublin, Ireland.

However, he also seems to have been fond of statistical fantasy worlds. This particular excursion brought back tangible and practical benefits. For this reason, his work was subsequently publicized and celebrated by a far more (in)famous man, Ronald Fisher.

Entering the fantasy world

So, in this fantasy world, we know the underlying normal distribution of samples

$$x \sim \mathcal{N}(\mu, \sigma)$$

and we are able to draw many independent sets of n samples,

$$x_{ki} = \{x_{k1}, \dots, x_{kn}\}$$

where i indexes samples and k indexes sets of samples with sample means $m_{i.}$ and

where r indexed samples and n indexed sets of samples, with sample means m_k and sample variances s_k^2

$$m_k = \frac{1}{n} \sum_i x_{ki}, \quad s_k^2 = \frac{1}{n} \sum_i (x_{ki} - \mu)^2,$$

The distribution of sample means m_k is

$$m_k \sim \mathcal{N}(\mu, \sigma/\sqrt{n})$$

but what is the distribution of sample variances s_k^2 ? To find the answer, we will need the **chi-square distribution**.

✓ Chi-square distribution

If $x \sim \mathcal{N}(0, 1)$ is a standard normal variable, then the distribution of the square $z = x^2$ is

$$z \sim \chi_1^2(z) = \frac{1}{\sqrt{2\pi z}} \exp\left(-\frac{z}{2}\right)$$

which is called the chi-square distribution with one degree of freedom. (No, we won't derive this, ask Grok3, if you must know.)

The distribution of a sum of independent squares $z = x_1^2 + \dots + x_n^2$

$$z \sim \chi_n^2(z) = \frac{z^{(n/2)-1} e^{-z/2}}{2^{n/2} \Gamma\left(\frac{n}{2}\right)}$$

is the chi-square distribution with n degrees of freedom, which also equals a Gamma distribution with shape parameter $n/2$ and rate parameter $1/2$. To see this, compare the definition of a Gamma distribution with shape parameter $\alpha = n/2$ and rate parameter $\lambda = 1/2$

$$P(x) = \frac{1}{\Gamma(\alpha) x} (\lambda x)^\alpha e^{-\lambda x}$$

So what can we say about the distribution of sample variance, s_k^2 , now that we have figured out the distribution of a sum of n independent squares, z ?

Dividing χ^2 samples z by $n - 1$ gives us the distribution of sample variance for a standard normal distribution, and scaling with σ^2 gives the same for a non-standard normal distribution.

$$s_k^2 = \frac{\sigma^2}{n-1} z, \quad z \sim \chi_{n-1}^2$$

where z follows a chi-square distribution with $n - 1$ degrees (not n degrees) of freedom. We

where z follows a chi square distribution with $n - 1$ degrees (not n degrees) of freedom. We use **Bessel's correction** because we had to estimate the mean before computing variance. Please use the 'toggle' in the code below to convince yourself that this improves the result noticeably, at least for small n !

Distribution of means and variances

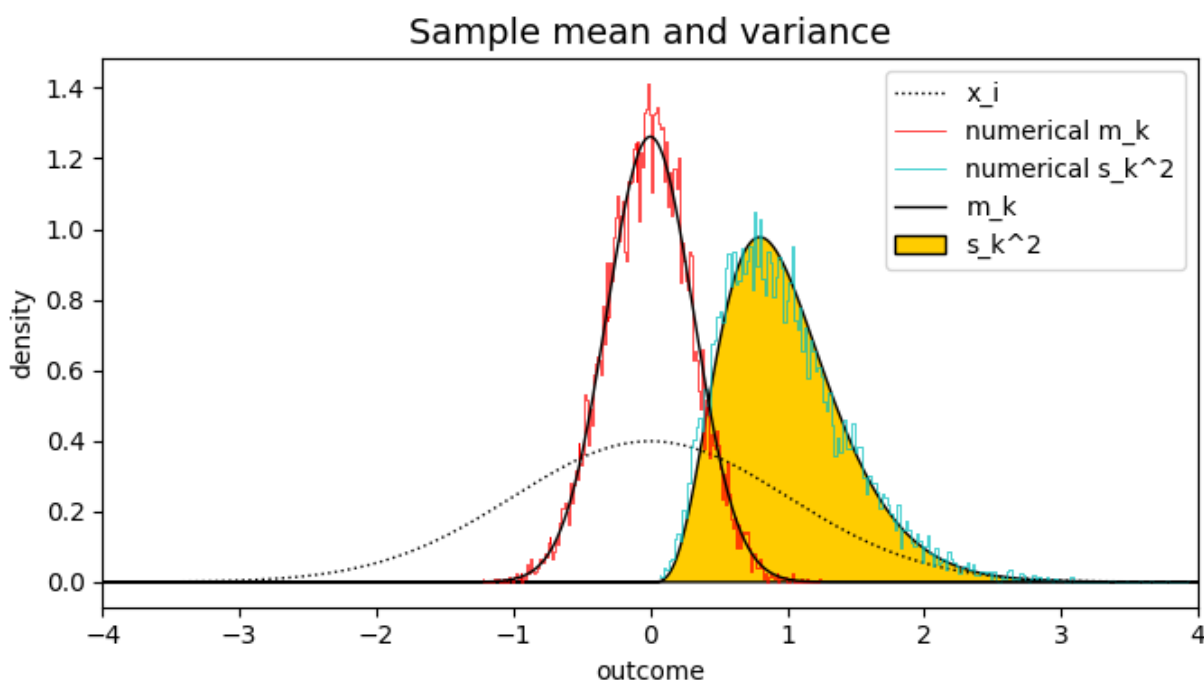
Show code

mu

sigma

n

toggle

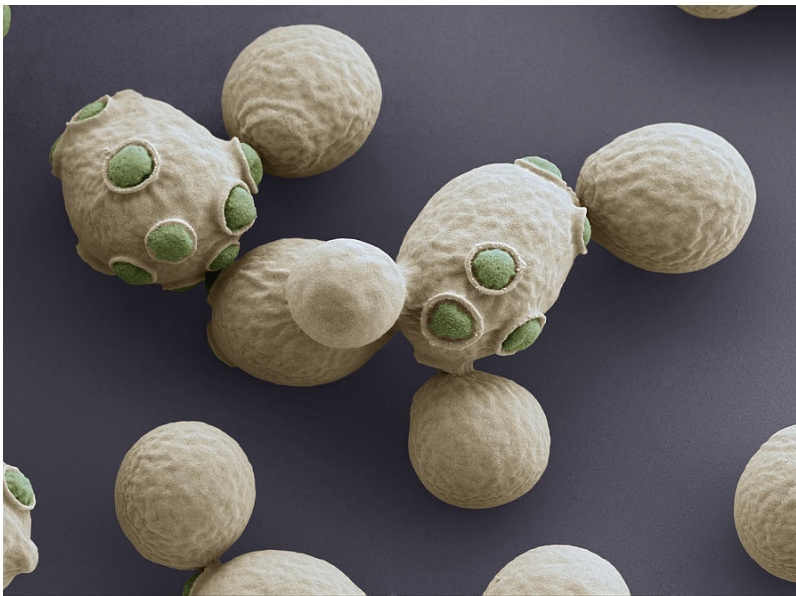


```
plot_distribution_means_variances
def plot_distribution_means_variances(mu=0, sigma=1.0, n=10,
toggle=0)
```

(Slide-25)

Returning to the brewery

Let's now return to the brewery, where Mr. Gosset wishes to estimate state of fermentation in a very large (and valuable) tank of Guinness by inspecting the yeast cells in a small number of samples:



Brewer's yeast

To estimate the distribution of the observed mean from n samples (how probable are different deviations from the true value?), he forms his famous **test statistic** t

$$t_m = \frac{m - \mu_0}{s^2 / \sqrt{n}}, \quad m = \frac{1}{n} \sum_i x_i, \quad s^2 = \frac{1}{n-1} \sum_i (x_i - m)^2$$

where μ_0 is a hypothetical true mean (e.g., a value to be ruled out or in).

While this is **not the z-score** of m that we would obtain from the true mean μ_0 and true standard deviation σ_0

$$z_m = \frac{m - \mu_0}{\sigma_0^2 / \sqrt{n}},$$

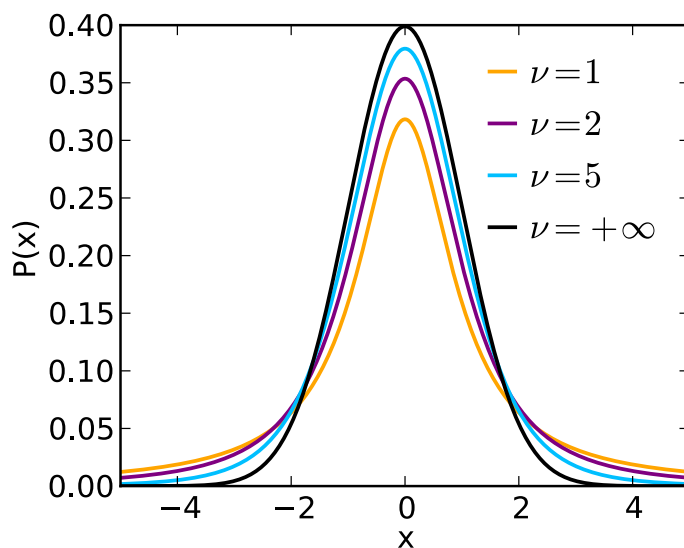
the idea is similar in that it serves to rescale our observations such that they approximate a standard normal distribution.

Instead, the t-statistic is the ratio of two random variables, m_k and s_k^2 , that are **uncorrelated** of each other. One way of seeing this uncorrelatedness is to note that m_k is a signed quantity (which depends on the sample signs, positive or negative), whereas s_k^2 is not (as it does not depend on sample signs).

The probability density of t_m is

$$f(t_m) = \frac{\Gamma\left(\frac{n}{2}\right)}{\sqrt{\pi(n-1)} \Gamma\left(\frac{n-1}{2}\right)} \left(1 + \frac{t_m^2}{n-1}\right)^{-\frac{n}{2}}$$

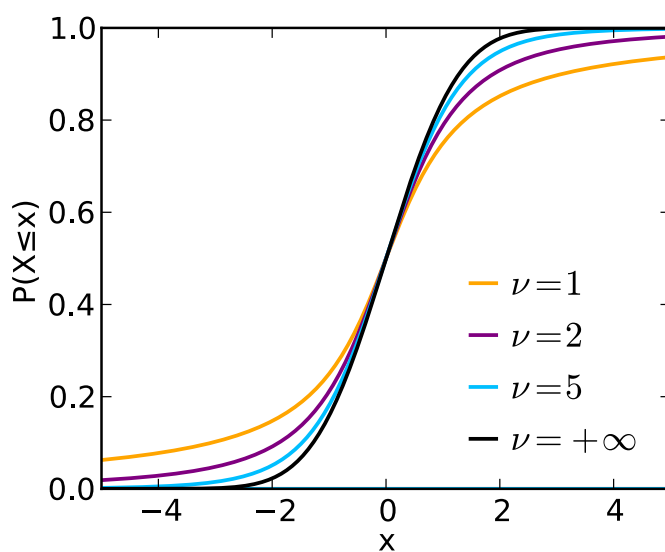
which has heavier tails than a standard normal distribution, reflecting extra uncertainty from using s instead of the true value σ .



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The **cumulative probabability** is defined as

$$F(t_m) = \int_{-\infty}^t f(t) dt$$



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In Python, you can import both functions as *t.pdf* and *t.cdf* from *scipy.stats*. The inverse of the cdf (probability percentile) is available as *t.ppf*.

In Matlab, you can use the built-in functions *tpdf()* and *tcdf()*. The inverse of the cdf is called *tinu()*.

Conclusion

- Student's t-distribution tells us how an observed sample mean is distributed around the true mean, **without knowing the true variance**.
- The distribution has heavier tails than a standard normal distribution, due to the extra uncertainty that comes from using the **sample variance**.
- The difference vanishes for large sample size, where sample variance approaches true variance.
- Given a sample size, we can use this to compute **confidence intervals** (see below).
- Given a desired level confidence, we can use it to compute the **required sample size**.
- For example, Mr. Gosset could use it to determine how many samples must be drawn from a tank of beer in order to be 99.99% certain that it meets the quality standards of Guinness!

4. Hypothesis testing

In statistical "hypothesis testing", we compare an observation x with a probability distribution $P(x)$ in order to determine the probability that a hypothesis is correct or incorrect.

Typical hypotheses are:

- Is the observed value **due to chance** and comparable to zero (null hypothesis H_0)?
- Is the observed value **not due to chance** and larger than zero (alternative hypothesis H_1)?

To make such determinations, we first have to choose a **level of statistical significance** α , which sets the acceptable probability of error. To get published in a ho-hum journal, we can use $\alpha = 0.05$ (or 5%), but for higher-impact journals we need $\alpha = 0.01$ or even $\alpha = 0.001$ (1% or 0.1%).

5% is considered 'significant', 1% 'highly significant', and 0.1% 'very highly significant'.

Next we convert our observation x into a probability value p , using probability distribution $P(x)$ (see below for details).

Lastly, we compare our **p-value** to the chosen α . For

$$p \leq \alpha$$

we report triumphantly that our finding is ((very) highly) significant, whereas for

$$p > \alpha$$

we concede reluctantly that our finding 'does not reach significance'. We would and should never call our precious results 'insignificant'.

By convention, we report our findings as $p < .05$, or $p < .01$, or *n. s.* (not significant). We **do not clutter up** the manuscript by writing something like $p = 0.0487$. Note further that, when reporting **p-values**, the zero before the decimal point is typically omitted: $p < .01$ rather than $p < 0.01$.

Normal distribution

In a normal distribution, $\sim 68.2\%$ of the probability mass falls within $\pm\sigma$ of the mean, $\sim 95.4\%$ within $\pm 2\sigma$, and $\sim 99.8\%$ within $\pm 3\sigma$ of the mean μ .

$$\int_{-1}^{+1} \Phi(x) dx \approx 0.682, \quad \int_{-2}^{+2} \Phi(x) dx \approx 0.954, \quad \int_{-3}^{+3} \Phi(x) dx \approx 0.998$$

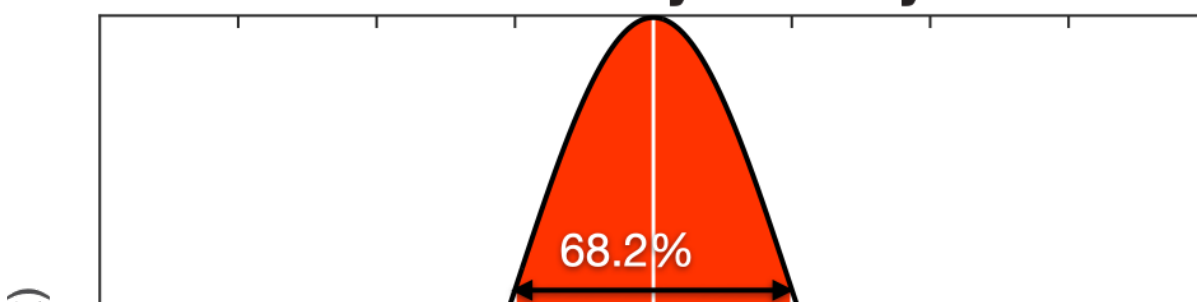
Thus, a significance level of $\alpha = 0.05$ (or 5%) implies that our observation falls with 95% probability **within** $\pm 2\sigma$ of the population mean, and with 5% probability **outside** this range.

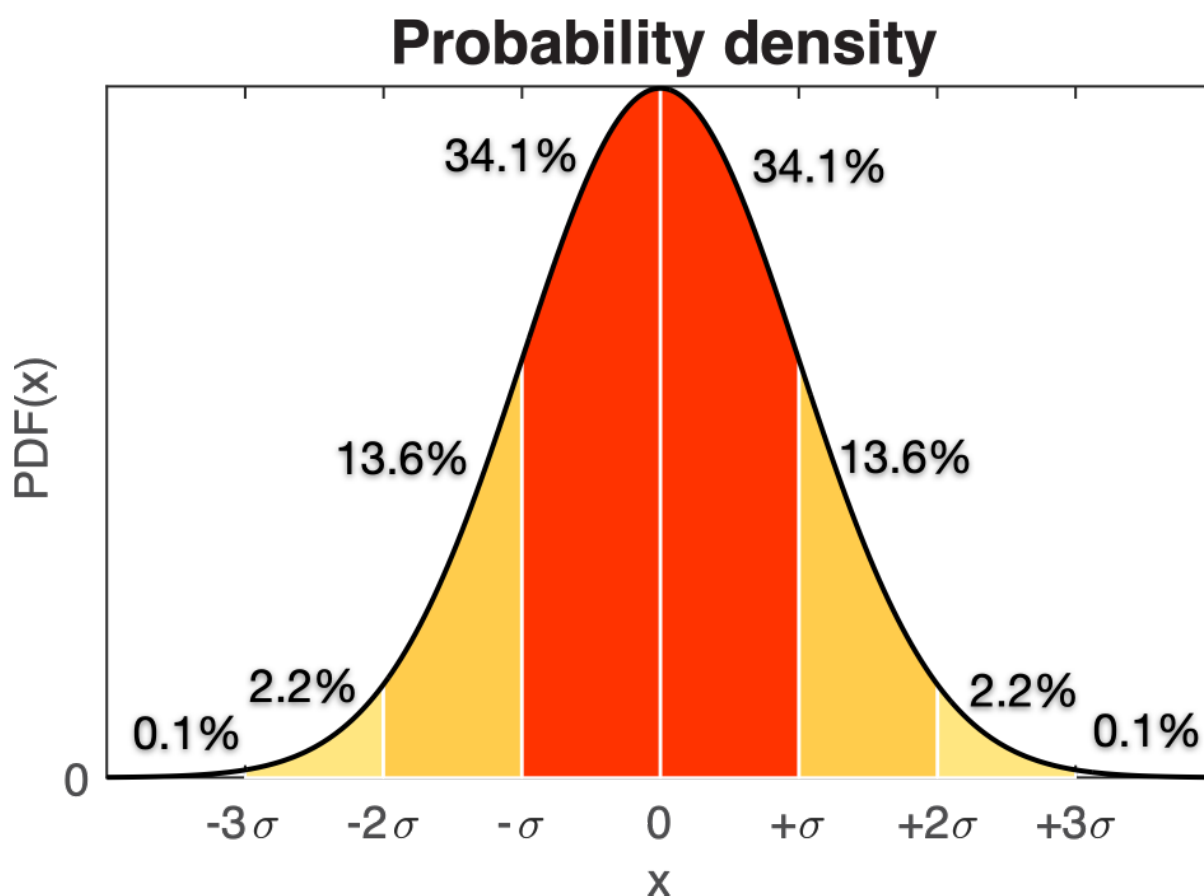
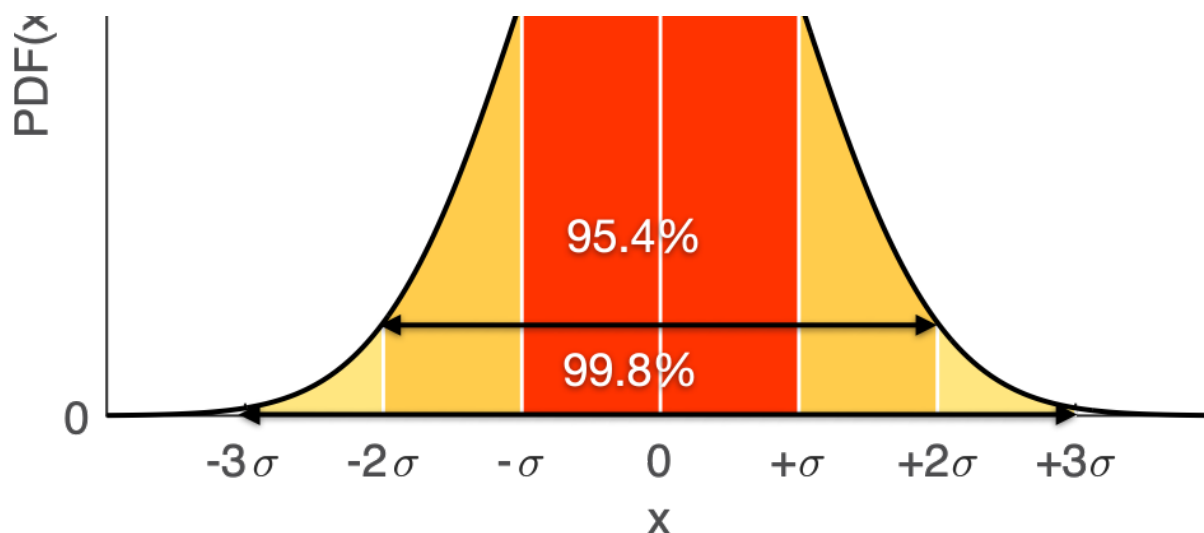
Similarly, a significance level of $\alpha = 0.001$ (or 0.1%) implies that our observation falls with 99.9% probability **within** $\pm 3\sigma$ of the population mean, and with 0.1% probability outside this larger range.

Another way of putting this that $\mu \pm 2\sigma$ is the 'two-sigma' confidence interval and $\mu \pm 3\sigma$ is the 'three-sigma' **confidence interval**.

All of this is intended to answer the question **how close?** (is an observation is from the distribution mean).

Probability density





P-values and Error Function Complement

If you are observing a standard normally distributed variable $z \sim \mathcal{N}(0, 1)$, the p-value that corresponds to an observed (absolute) value $|z|$ is

$$p = \text{erfc} \left(\frac{|z|}{\sqrt{2}} \right) = \frac{2}{\sqrt{2\pi}} \int_{|z|}^{\infty} e^{-z^2/2} dz$$

Note that this p -value includes the probability mass of **both tails**. For the one-sided probability, you need to divide by 2.

Both Python and Matlab provide built-in functions *erfc*

For Python you need to import

from scipy.special import erfc

✓ Cumulative normal distribution

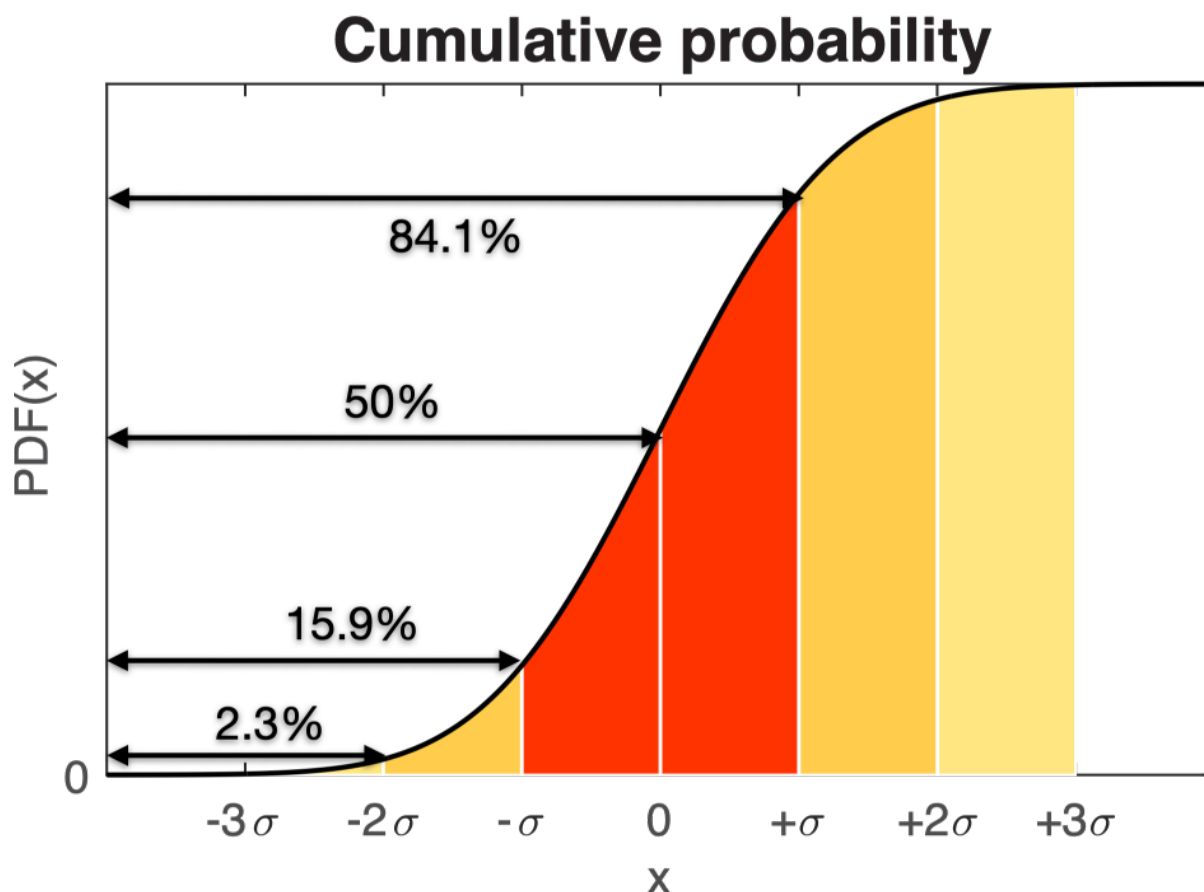
If instead we want to answer the question **how far?** (is an observation is from the distribution mean), we use the cumulative normal distribution.

Here, $\sim 2.3\%$ of the probability mass fall **below** -2σ of the mean, $\sim 15.9\%$ fall **below** $-\sigma$, and 50% below the mean.

$$\int_{-\infty}^{-2} \Phi(x) dx \approx 0.023, \quad \int_{-\infty}^{-1} \Phi(x) dx \approx 0.159, \quad \int_{-\infty}^0 \Phi(x) dx = 0.5$$

For example, if our observed value is $x_{obs} = -2\sigma$, it falls into the lowest 2.5% of the probability mass. Accordingly, we can report $p < .05$ ('significant').

Similarly, if our observed value is $x_{obs} = -3\sigma$, it falls into the lowest 0.1% of the probability mass. Accordingly, we can report $p < .001$ (very highly significant).

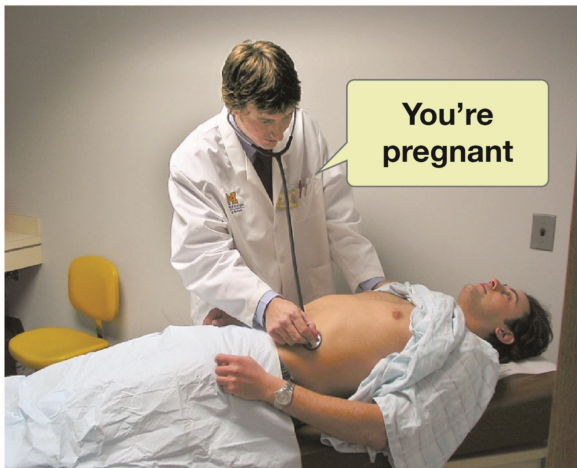


Double-click (or enter) to edit

Type I and type II errors

As no statistical inference is perfect, we are bound to make errors. One error is to **mistakenly fail to reject** a hypothesis that is actually false (false positive, Type I error), another is to **mistakenly reject** a hypothesis that is actually true (false negative, Type II error).

Type I error
(false positive)



Type II error
(false negative)



@Michael Katsillis

Conclusion

- The drawing and reporting of statistical conclusions follows certain conventions.
- First, we choose confidence level α **as low as** our results allow (0.05 or 0.01 or 0.001). The lower the better!
- Next, we choose expected distribution $P(x)$ for our observations. This choice depends on the question we are asking. The expected distribution can come from a null hypothesis, from our favourite hypothesis, from another experiment, from bootstrapping, or many other sources.
- Finally, we compare our observation x to the distribution $P(x)$. Does it fall near the mean or far from the mean? Specifically, does it fall **inside or outside** the confidence intervals for $P(x)$ specified by α ?
- We report the p-value of the hypothesis that we **exclude**. We can **only disprove**, never prove! This is why **small** p-values give us **high** statistical confidence.

✓ 5. Student's t-test for different mean

First we consider the *Single-Sample t-Test*, which checks whether the mean m of an observed sample differs significantly from a hypothetical population mean μ_0 . Often, this hypothetical mean is $\mu_0 = 0$, so that we check whether the observed mean differs significantly from zero.

Given a sample $x_i = \{x_1, \dots, x_n\}$ with n values, we compute the mean, m , the corrected standard deviation, s , and the t-statistic, t :

$$m = \frac{1}{n} \sum_i x_i, \quad s = \sqrt{\frac{1}{n-1} \sum_i (x_i - m)^2}, \quad t = \frac{m - \mu_0}{s/\sqrt{n}}$$

Next we compare the t-statistic to the Student's t-distribution with $n - 1$ degrees of freedom, as illustrated in the code below. Specifically, we compare the observed value t_{obs} to the quantiles that represent our chosen confidence bounds, for example, $q_{low} = 0.025$ and $q_{high} = 0.975$ for $\alpha = 0.05$.

Quantiles are obtained from the inverse cumulative distribution, which is also called the percent-point function (PPF).

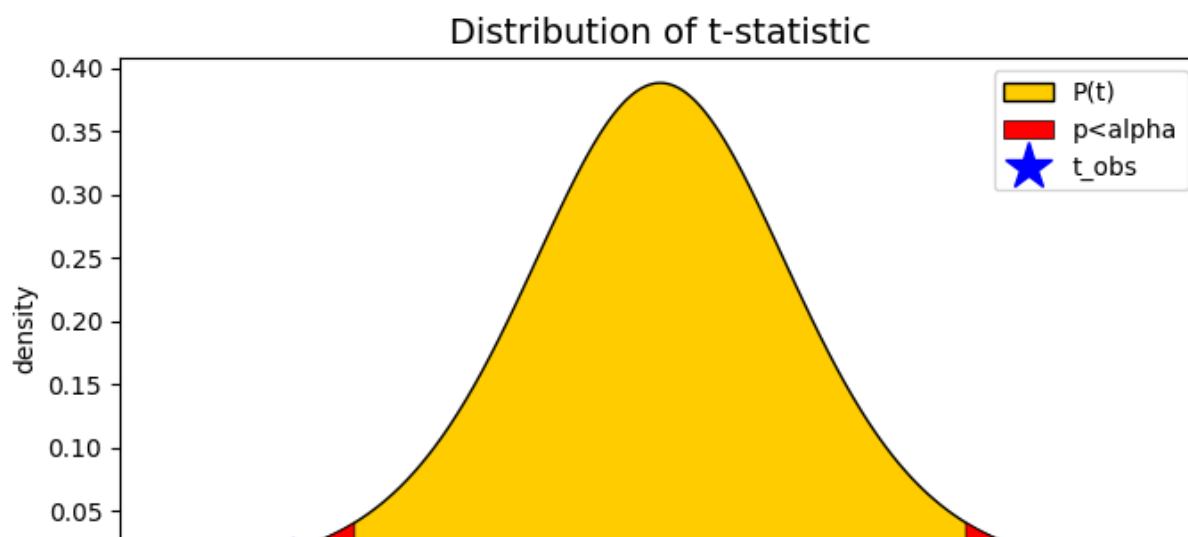
In Python, you can import this as *t.ppf* from *scipy.stats*.

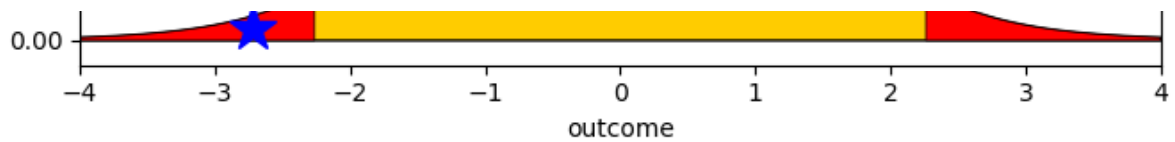
In Matlab, you can use the built-in functions *tinvt*.

Single-Sample t-Test

Show code

alpha 0.05
 n 10
 toggle 0





```
plot_single_sample_t_test
```

```
def plot_single_sample_t_test(alpha=0.05, n=10, toggle=0)
```

(Slide-25)

Independent (unpaired) t-test

This applies to situations in which we have two independent (unpaired) samples, for example observations x_{1i} from n_1 wild-type mice and observations x_{2i} from n_2 mutant mice.

Here we would like to know whether our sample means differ significantly?

The sample means are

$$m_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} x_{1i}, \quad m_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} x_{2i}$$

and the estimated variances of **individual samples** are

$$s_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (x_{1i} - m_1)^2, \quad s_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (x_{2i} - m_2)^2$$

and the estimated variances of **sample means** (i.e., the squared standard errors) and of the **difference between sample means** are

$$\frac{s_1^2}{n_1}, \quad \frac{s_2^2}{n_2}, \quad \frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}$$

The standard error of the difference is therefore

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

and the test statistic for the difference is

$$t_{diff} = \frac{(m_1 - m_2) - (\mu_1 - \mu_2)}{SE}$$

For a null hypothesis $\mu_1 = \mu_2$ of zero difference, the test statistic becomes

$$t_{diff} = \frac{m_1 - m_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

with $n_1 + n_2 - 2$ degrees of freedom.

Paired t-test

This applies to situations in which we have paired samples, for example observations before and after treatment/lesion of the same n individuals

Again we would like to know whether our sample means differ significantly?

The average difference between sample pairs is

$$m_D = \frac{1}{n} \sum_{i=1}^n d_i, \quad d_i = x_{1i} - x_{2i}$$

and the estimated variance of the **differences**

$$s_D^2 = \frac{1}{n-1} \sum_{i=1}^n (d_i - m_D)^2$$

The standard error of **differences** is

$$SE_D = \frac{s_D}{\sqrt{n}}$$

and the test statistic for **differences** is

$$t_{diff} = \frac{m_D}{s_D / \sqrt{n}}$$

with $n - 1$ degrees of freedom.

Summary

- t-tests are useful for assessing the significance of observed sample means **when the number of samples is small**.
- For large sample numbers, the normal distribution will do.
- The **single-sample** version tests for a significant difference from a null hypothesis (e.g., zero mean).
- The **unpaired** version tests for a significant difference between the means of two independent samples.
- The **paired** version tests for a significant difference between means when the individual samples are paired (e.g., derive from the same animal before and after treatment).

✓ 6. F-Test for different variance

When two distributions have similar means (i.e., the difference is not significant), they might nevertheless have different variances.

To test for differences in variance, William Gosset's approach was extended by Ronald Fisher as follows:

We have two independent samples of sizes n_1 and n_2 , with variances and means

$$s_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (x_{1i} - m_1)^2, \quad m_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} x_{1i}$$

$$s_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (x_{2i} - m_2)^2, \quad m_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} x_{2i}$$

Assuming that the underlying distributions are normal, with (unknown) variances σ_1^2 and σ_2^2 , we expect the sample variances (suitably rescaled) to follow χ^2 distributions:

$$\frac{n_1 - 1}{\sigma_1^2} s_1^2 \sim \chi_{n_1-1}^2$$

$$\frac{n_2 - 1}{\sigma_2^2} s_2^2 \sim \chi_{n_2-1}^2$$

As Fisher was an exceptionally vain and arrogant man, it is no surprise that he called his statistic the 'F-statistic'. The F-statistic is the ratio of the two sample variances

$$f = \frac{s_1^2}{s_2^2}$$

Under the null hypothesis of identical variance, $\sigma_1^2 = \sigma_2^2$, we can express this in terms of χ^2 samples

$$f = \frac{\frac{(n_1-1)s_1^2}{\sigma_1^2} \cdot \frac{1}{n_1-1}}{\frac{(n_2-1)s_2^2}{\sigma_2^2} \cdot \frac{1}{n_2-1}} \sim \frac{\frac{\chi_{n_1-1}^2}{n_1-1}}{\frac{\chi_{n_2-1}^2}{n_2-1}}$$

This distribution of f is called an F-distribution with $(n_1 - 1, n_2 - 1)$ degrees of freedom and can be computed from a regularized incomplete beta function:

$$F(f, d_1, d_2) = I_{\frac{fd_1}{fd_1+d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$$

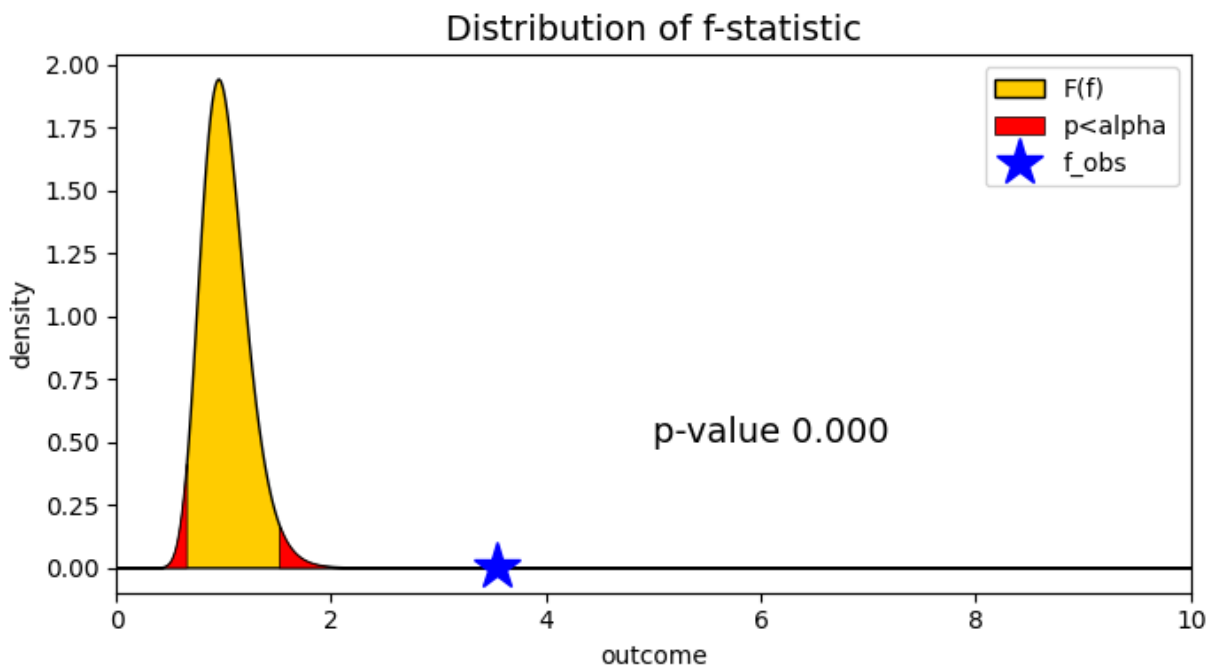
In Python, you can compute this distribution by importing *f.pdf* or *f.cdf* or *f.pmf* from stats.

In Matlab, you can use the built-in functions $fpdf(f, d1, d2)$ or $fcdf(f, d1, d2)$.

F-test for equal variance

Show code

alpha	<input type="range" value="0.05"/>	0.05
n1	<input type="range" value="92"/>	92
sigma1	<input type="range" value="4"/>	4
n2	<input type="range" value="92"/>	92
sigma2	<input type="range" value="7"/>	7
toggle	<input type="range" value="0"/>	0



```
plot_f_test_for_variance
def plot_f_test_for_variance(alpha=0.05, n1=10, sigma1=2, n2=8,
sigma2=4, toggle=0)
```

(Slide-25)

Summary

- The f-statistic tests for different variance in two samples.
- This will become relevant in the context of 'analysis of variance' (ANOVA).

7. Chi-Square Test for different distributions

Let's raise our ambition and ask a larger question: are two sample sets drawn from the same underlying distribution, or from two different distributions?

In proper statistical language: "Can we disprove, to a certain level of significance, the null hypothesis that the two data sets stem from the same underlying distribution?"

Asking whether two distributions are different is quite common in research. However, this question can arise in different contexts: (1) Are our observations continuous or binned (in discrete categories)? (2) Are we comparing one data sample to a known reference distribution, or are we comparing data samples from two equally unknown distributions?

The accepted test for differences between **binned data sets** is the **Chi-square test**.

If your data is continuous, you can bin it, of course, but this would incur a loss of information. For this reason, continuous data is typically assessed with the **Kolmogorov-Smirnov test (KS test)**.

Chi-square comparison to a known distribution

Let's assume that our observations fall into N distinct categories (i.e., they are binned) so that they can be summarized in terms of integer counts:

$$n_i \in \mathbb{N}, \quad i \in \{1, \dots, N\}, \quad N_n = \sum_i n_i$$

and that we have a null hypothesis about the number of observations that we expect in each bin:

$$e_i \in \mathbb{R}, \quad i \in \{1, \dots, N\}, \quad N_e = \sum_i e_i$$

The chi-square statistic is defined as

$$\chi^2 = \sum_{i=1}^N \frac{(n_i - e_i)^2}{e_i}$$

and samples a χ^2 -distributed with N degrees of freedom

$$\sum \frac{(n_i - e_i)^2}{e_i} \sim \chi_N^2$$

$i \quad \cup \quad i$

assuming that the number of model events N_e and observed events N_n are independent. When the model events are renormalized so that $N_e = N_n$, there are only $N - 1$ degrees of freedom.

The χ^2 -statistic follows a Gamma distribution with shape parameter $N/2$ and scale parameter $\chi^2/2$. The cumulative distribution is an incomplete Gamma function with the same arguments.

The code below uses the `scipy` functions *gamma.pdf* for density, *gamma.cdf* for cumulative probability, and *gamma.ppf* for quantiles.

Motivation of chi-square statistic

What motivates the odd form of the chi-square statistic? Under the null-hypothesis, the n_i are binomially distributed

$$n_i \sim \mathcal{B}(N_{tot}, p_i), \quad p_i = \frac{e_i}{N_{tot}}$$

with an expected mean and variance of

$$\langle n_i - e_i \rangle = 0, \quad \langle (n_i - e_i)^2 \rangle = N_{tot} p_i (1 - p_i) = e_i \left(1 - \frac{e_i}{N_{tot}} \right) \approx e_i$$

In other words, e_i approximates the variance of $n_i - e_i$, so that the square root of the fractional terms are approximately normally distributed

$$\frac{(n_i - e_i)}{\sqrt{e_i}} \sim \mathcal{N}(0, 1)$$

and the entire sum is χ^2 -distributed with N degrees of freedom

$$\sum_i \frac{(n_i - e_i)^2}{e_i} \sim \chi_N^2$$

The approximation gets better when the number of bins or the number of events per bin are large.

- ✓ Chi-square comparison of two unknown distributions

When we compare two (equally binned) data sets from unknown distributions,

$$r_i \in \mathbb{N}, \quad i \in \{1, \dots, N\}, \quad N_r = \sum_i r_i$$

$$s_i \in \mathbb{N}, \quad i \in \{1, \dots, IV\}, \quad IV_s = \sum_i s_i$$

the chi-square statistic becomes

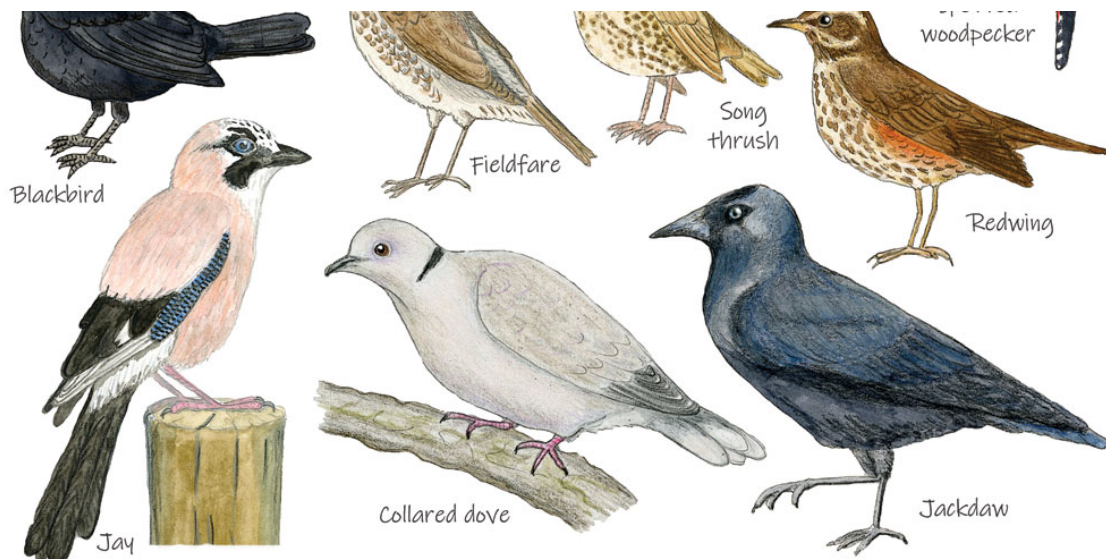
$$\chi^2 = \sum_{i=1}^N \frac{(r_i - s_i)^2}{r_i + s_i}$$

Note that r_i and s_i in the denominator approximate the **variance** of r_i and s_i , respectively. They are summed in the denominator because the variance of a difference is the sum of the individual variances.

The number of the degrees of freedom is N (the number of bins), provided that N_r and N_s are allowed to differ. If they are constrained to be equal, it is $N - 1$.

Explanation from PTVF 2002: "A birdwatcher wants to know whether the distribution of sighted birds as a function of species is the same this year as last. Each bin corresponds to one species. If the birdwatcher takes his data to be the first 1000 birds that he saw in each year, then the number of degrees of freedom is $N-1$. If he takes his data to be all the birds he saw (on specific days of both years), then the number of degrees of freedom is N . In the latter case, he is also testing whether the birds were more numerous overall in one year or the other: that is the extra degree of freedom."





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Chi-square comparison of two unknown distributions

Show code

alpha 0.05

Nbin 10

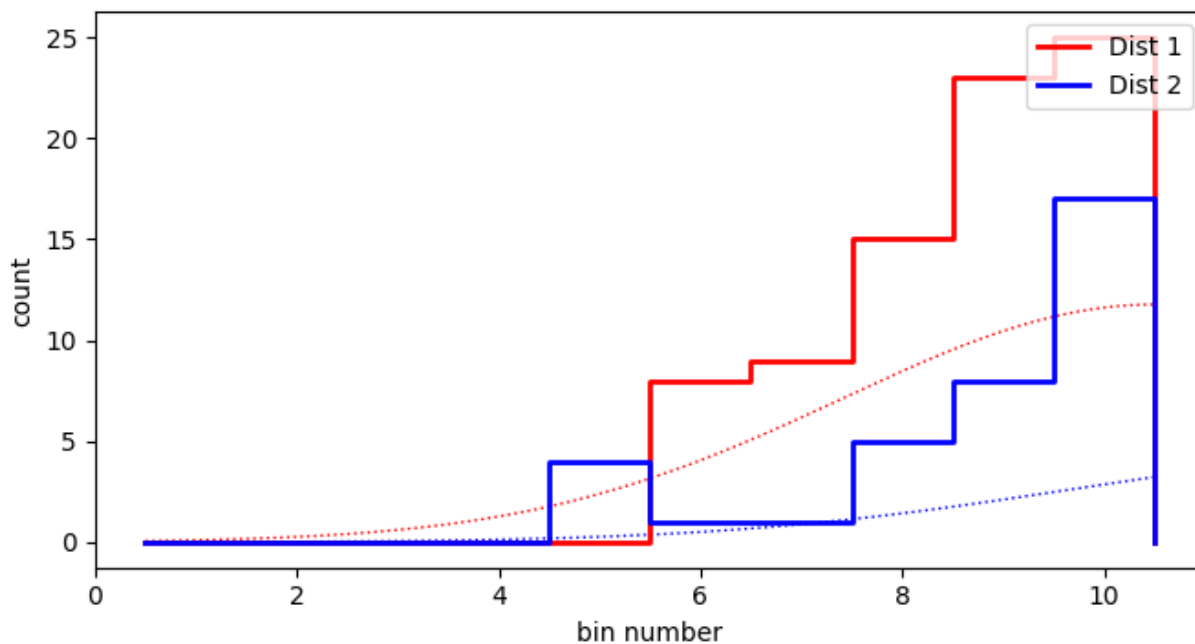
Ntot 200

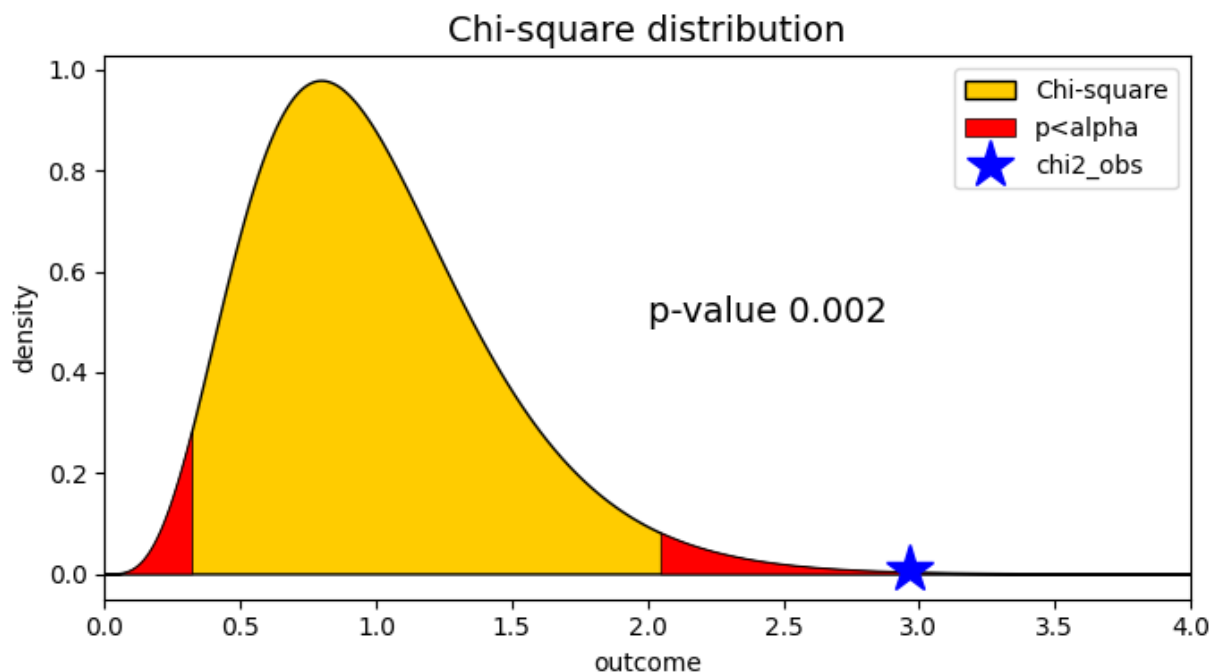
p1 0.05

p2 0.06

toggle 0

Two count distributions over bins





```
plot_chi_square_compare
def plot_chi_square_compare(alpha=0.05, Nbin=10, Ntot=100, p1=0.05,
                             p2=0.06, toggle=0)
```

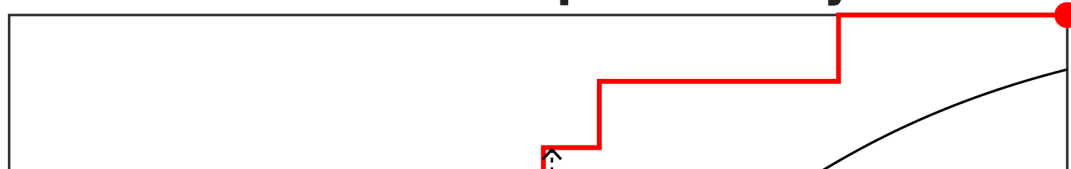
(Slide-25)

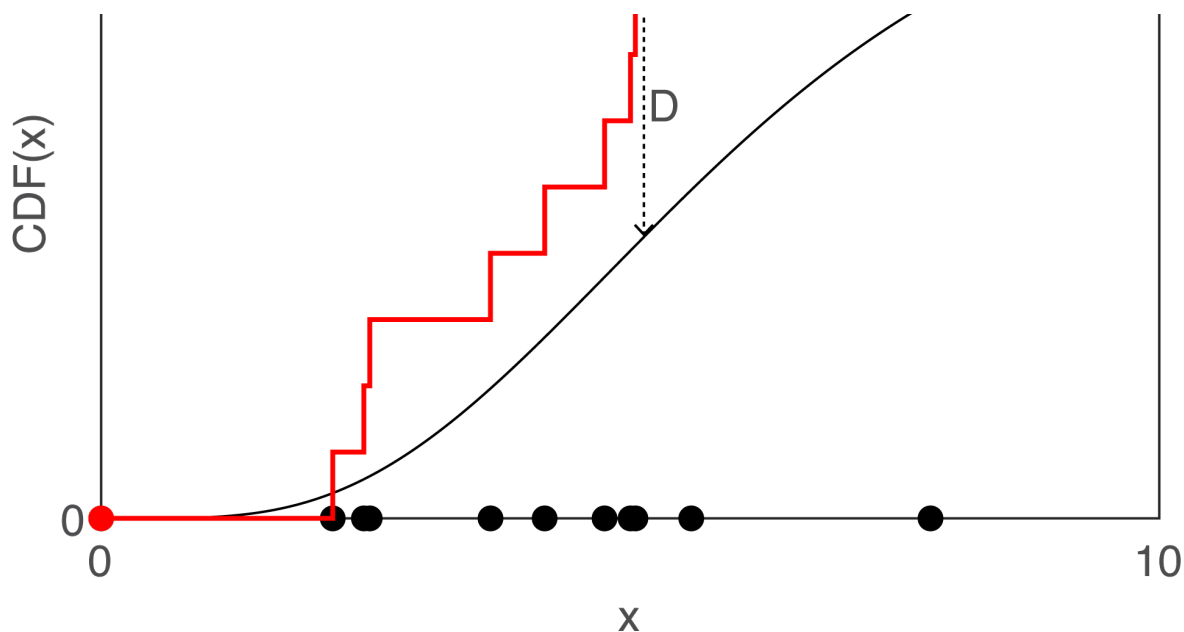
8. Kolmogorov-Smirnov (KS) test for different distributions

We conclude this lecture with a test that is **non-parametric** in the sense that it does not assume normally distributed observations, but that does rely on an analytical distribution. The KS test is a widely accepted tool for assessing the difference between distributions.

The KS test uses cumulative probability functions (CDFs). Suppose you have observed N data points x_i , with $i \in \{1, 2, \dots, N\}$. The CDF is an integer function $S(x)$ counting the fraction of data points to the left of x . It looks like a staircase and increments by $1/N$ at every data point x_i .

Cumulative probability





While one can imagine many measures for the difference between CDFs, the Kolmogorov-Smirnov measure is particularly simple. It is defined as the absolute value of the **maximal distance** between two CDFs.

When we compare the observed $S_N(x)$ in our N samples to a known CDF $P(x)$, the K-S statistic is

$$D = \max_x |S_N(x) - P(x)|$$

whereas when comparing two observed functions $S_{N_1}(x)$ and $S_{N_2}(x)$, the K-S statistic is

$$D = \max_x |S_{N_1}(x) - S_{N_2}(x)|$$

The function for computing significance is the monotonically decreasing function

$$Q_{KS}(\lambda) = 2 \sum_{i=1}^{\infty} (-1)^{i-1} e^{-2i^2 \lambda^2}$$

The significance level (p-value) of an observed value of D (which can disprove the null hypothesis) is given by

$$\text{Probability}(D > D_{obs}) \approx Q_{KS} \left[\left(\sqrt{N_e} + 0.12 + 0.11/\sqrt{N_e} \right) D_{obs} \right]$$

where N_e is the degrees of freedom. When we have one data set with N samples, we have $N_e = N$. When we have two data sets with N_1 and N_2 samples, then

$$N_e = \frac{N_1 N_2}{N_1 + N_2}$$

$$N_1 + N_2$$

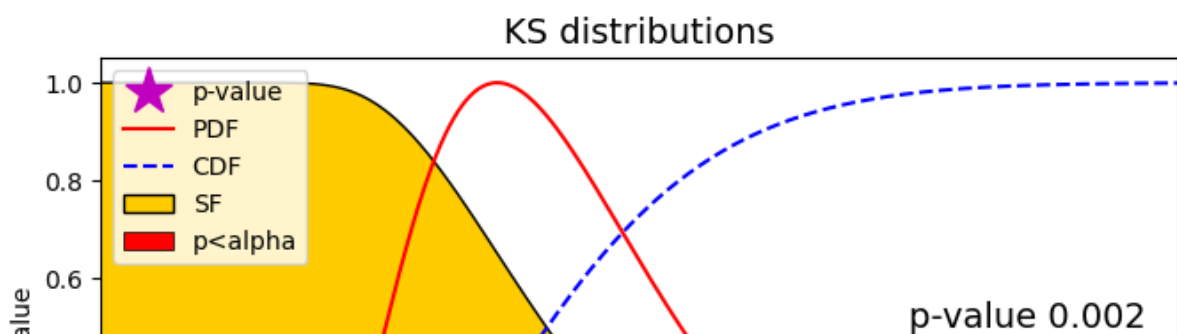
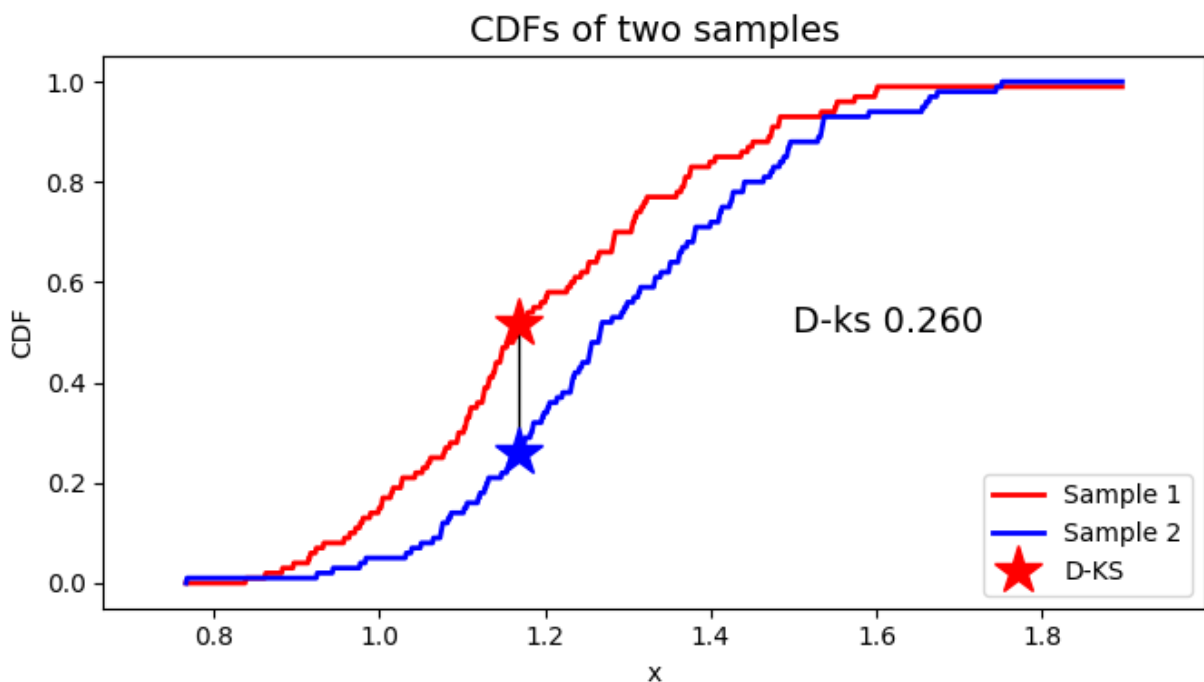
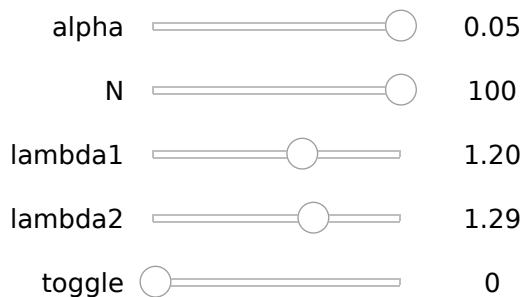
Note that sensitivity of the KS-test is not independent of x , because the distribution of deviations from $S(x)$ is not independent of x : large deviations are more likely in the intermediate region $S(x) \approx 0.5$ than at either extreme, $S(x) \approx 0$ or $S(x) \approx 1$.

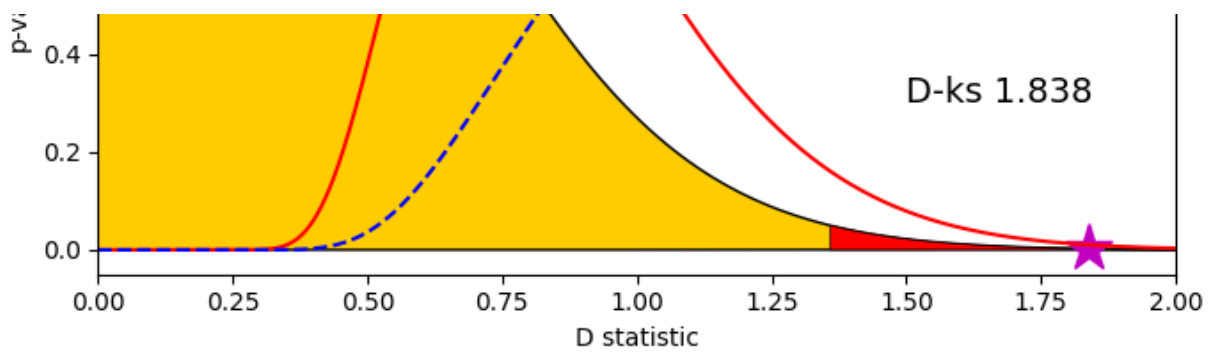
In Python, you can import `kstest` or `ks_2samp` from `scipy.stats`.

In Matlab, you can use built-in function `kstest` or `kstest2`.

KS-test for two samples

Show code





plot_ks_two_samples

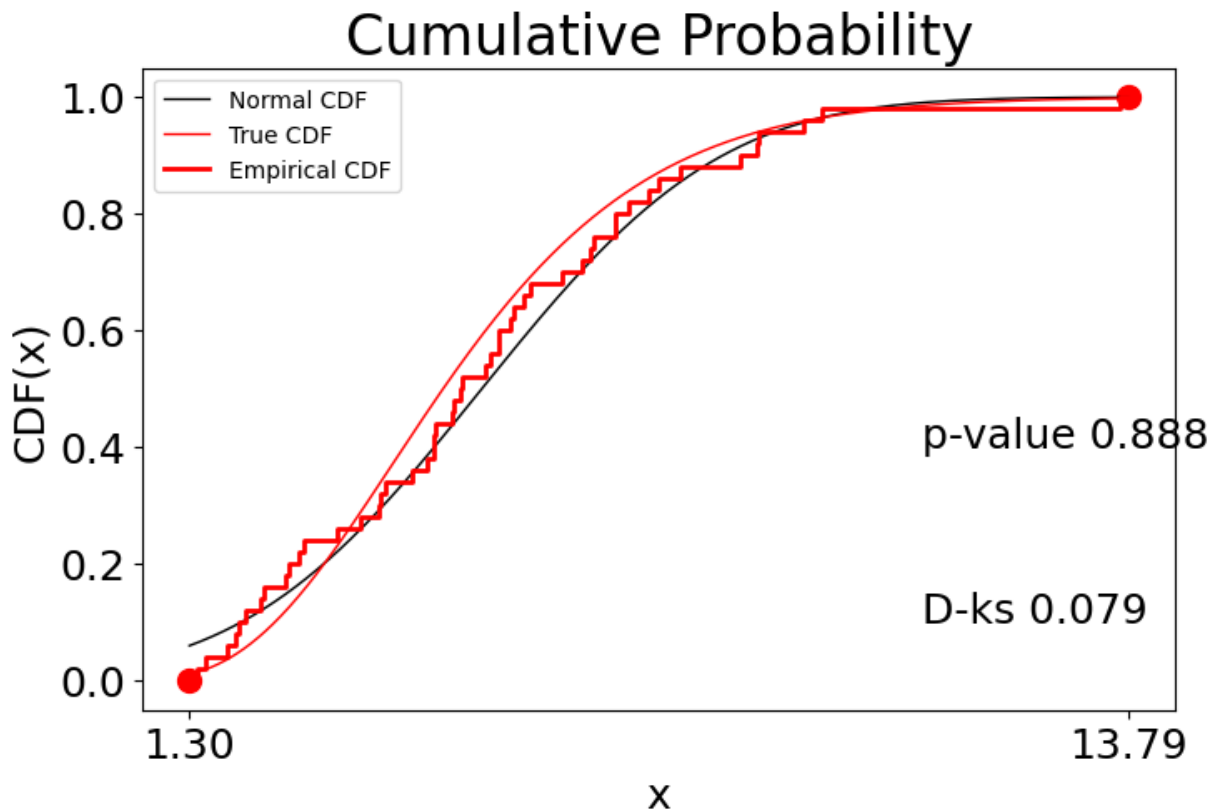
```
def plot_ks_two_samples(alpha=0.05, N=10, lambda1=1, lambda2=1.2,
toggle=0)
```

(Slide-25)

KS-test for normality

Show code

a 5
N 50
toggle 0



plot_ks_test_for_normality

```
def plot_ks_test_for_normality(a=5, N=50, toggle=0)
```

(Slide-25)
